

[ASRJC/2024/Q2]

2 Name your Jupyter Notebook as:

TASK2_<your name>_<centre number>_<index number>.ipynb

Knowing how important data structures are in real life tech interviews in assessing problem solving skills, a Computing student decides to implement a hybrid data structure integrating binary search tree, linked list, stack and queue to manage players' game scores.

Implement a binary search tree (BST) of stack or queue. Each BST node comprises the following:

- `ds`: a variable having the value 'S' (stack) or 'Q' (queue)
- `score`: game score
- `data`: stores player ids with the same score
 - if `ds` is 'S', `data` is a stack implemented using linked list
 - if `ds` is 'Q', `data` is a wraparound queue implemented using array
- `left`: pointer to left subtree
- `right`: pointer to right subtree

You may assume that a player only has one game score.

Each stack or queue can store a maximum of 10 player ids with the same game score.

- a stack has a `top` pointer where insertion (`push`) and deletion (`pop`) happen, as well as a `size` attribute representing the number of items in the stack
- a queue has a `front` pointer where deletion (`dequeue`) happens and a `rear` pointer where insertion (`enqueue`) happens, as well as a `size` attribute representing the number of items in the queue

Each data structure class has the following methods:

- `BSTNode`: constructor/initialiser
- `BST`: constructor/initialiser, `insert()`, `search()`, `inorder()`
- `LinkedListNode`: constructor/initialiser
- `LinkedList`: constructor/initialiser, `insert()`
- `Stack`: constructor/initialiser, `push()`, `pop()`, `display()`
- `Queue`: constructor/initialiser, `enqueue`, `dequeue`, `display()`

Task 2.1

Write program code to create the classes `BSTNode`, `BST`, `LinkedListNode`, `LinkedList`, `Stack` and `Queue` with the associated attributes and methods. [16]

Task 2.2

Write program code to insert data from `scores.txt` into the `BST` using the `insert()` method. Verify that the data has been inserted successfully using the `inorder()` method. [4]

Task 2.3

Write program code to search for the scores 75 and 53. Output

- the player id(s) for a successful search
- "Not found" for an unsuccessful search

[3]

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Task 2.4

Write program code to implement and test a `maximum()` method to determine the highest score and output the player id(s) with this score. [4]

Save your Jupyter Notebook for Task 2.